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# Data Driven based Malicious URL Detection using Explainable AI

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***Abstract*—With the ever-increasing reach of the internet, and its increasing access through various types of devices, the spread of malware, phishing attempts, etc. have steadily been increasing, along with their level of sophistication. Thus it becomes very important to conduct research on different methods to prevent such harmful attacks on systems and users. Using a malicious URL is the common way for hackers to attack a system, thus, to accommodate the variety attack vectors of malicious websites, 21 features were extracted from 651,191 URLs to train the proposed model. A two-stage stacked ensemble learning model, based on gradient boosting methods and random forest, has been trained and tested in the 70:30 ratio of the 651,191 URLs, and an accuracy of 97% has been achieved. Then Explainable AI (XAI) has been used to clearly explain the working of the model, and study the impact of each of the 21 features on the 4 class predictions (benign, defacement, phishing and malware).**

***Index Terms*—malicious URL detection, ensemble-learning, random forest, gradient boosting, explainable AI**

## I. Introduction

The advancement of technology and communications has made our lives infinitely easier, a prime example being the World Wide Web (WWW). Ever since the advent of the internet and the WWW, access to information about almost everything has given mankind innumerable opportunities, and non-exhaustive knowledge and led to unimaginable progress. New communication technologies have led to the growth of new businesses such as online banking [1], personalized advertising, social media networks, and e-commerce to name a few. Having an online presence, nowadays, has become necessary for both the consumer and the business. The improvement of mobile technology helped the internet to get a much greater reach, increasing the number of internet users everywhere.

However, with the ever-increasing reach of the internet, and its increasing access through various types of digital, and mobile devices, the spread of malware, computer viruses, phishing attempts, etc. has been steadily increasing. With the improvement of technology, the methods of attacks and online scams are getting more sophisticated as well. Such attacks may include links that upon clicking get the user's sensitive information, which may lead to financial or identity theft. Attacks may also install malware or spyware in a user's system. Phishing [2], drive-by downloads [3], social engineering [4], SQL injections [5], etc. are just few of the attacking methods. Spreading corrupted URLs, however, is the most common form of attack, so efficient techniques must be developed to counter this breach.

A URL or Uniform Resource Locator [6], is a global address of documents and resources on the WWW, and malicious URLs are those that have been compromised and are used by harmful entities to redirect users to unwanted or phishing sites, malware downloads, etc. Two basic components of the URL are protocol identifier - indicating which protocol the site uses, and resource name - identifying the IP address or domain name [7]. Hackers can easily alter one or more components of a URL to attack systems, and they can easily be spread through files and shared networks. According to Symantec's Internet Security Threat Report, in 2019, using malicious URLs for attacking systems was ranked first among the top 10 common attack methods.

To counter this form of attack, web/cyber security services have developed a list of blacklisted sites, which are marked harmful by honeypots, web crawlers combined with site analysis heuristics [8], or through crowdsourcing. However, to bypass this method, hackers can easily make minute changes to the URL string and hence deceive security. So, the traditional method of blacklisting malicious URLs is non-exhaustive, and results in a high number of false negatives [7]. New corrupted URLs cannot be added in time to the list as they are too recent, or were incorrectly marked [9]. Since it is hard to keep track of the constantly increasing number of corrupted URLs, it becomes important to overcome the limitations of traditional security techniques to detect and prevent these malicious URLs.

Previous research works classify URLs as safe and unsafe, and usage of Explainable AI has not been demonstrated in this domain before. There are works using machine learning and/or deep learning for classification, This work proposes a multi-class classification model, which is based on an ensemble learning-based approach to classifying URLs into 4 categories - benign, defacement, phishing, and malware, based on the different features/components of the URL string. To train and test the model, a dataset of 651,191 URLs was used. 21 features were extracted from the input URL strings - abnormal URL, count of special characters such as '.', '@', '?', etc., URL length, count of digits, etc. Then the data is fed into a stacked ensemble machine learning model. To explain the

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working of the ensemble model, Explainable AI (XAI) is used, to show the contribution of each feature on the model prediction results.

### A. Our Contributions

The main contributions of this paper are:

- A malicious URLs dataset of 651,191 URLs is used in a 70:30 ratio for training and testing the model respectively.
- Multi-class classification is performed - machine learning and deep learning models are used to classify URLs as benign (or safe), defacement, phishing, or malware based on an input of 21 features.
- The proposed multi-class classification model is a stacked ensemble machine learning model based on gradient boosting methods and random forest, and accuracy of 97% is achieved.
- Further, Explainable AI (XAI) is used to show the impact of each of the 21 features on the output classification result.

The paper is structured as follows: Section 2 consists of related works. Section 3 gives the proposed methodology. Sections 4 and 5 gives the performance evaluation, and conclusions ad future works respectively.

## II. Related Works

Detecting malicious URLs is a major topic of concern for cybersecurity experts. Several approaches and methods have been proposed for detecting unsafe URLs to protect systems and their users from getting attacked. These approaches include classifying sites as safe or unsafe based on blacklist-based detection or content-based detection. Even though the centralized blacklist method is the most used detection method in practice, blacklists fail to maintain an exhaustive list of malicious URLs, as new unsafe URLs are generated easily and very frequently [9] algorithmically by hackers, which makes it impossible to detect them. Another detection method is web-content-based detection, in which the HTML code, JavaScript features, etc. is processed [10]. However, some malware can be downloaded while crawling the content of an infected website, and the security of the system can be compromised.

Raja et al. [11] conducted research on extracting lexical features from a dataset containing 68,851 malicious URLs, and using various models - support vector classifier, logistic regression, k-nearest neighbours, Naive Bayes, and Random Forest, to classify URLs as malicious or benign. The Random Forest model gave the highest accuracy of 99%. Patil et al. [12] conducted research for binary classification of URLs (safe or unsafe). The authors used 6 decision tree (DT) based algorithms - J48 DT, SimpleCart, Random Forest, Random Tree, ADTree, and REPTree, to classify 52,082 URLs as malicious or benign. Ghaleb et al. [13] proposed an ensemble learning model using Random Forest as the base model and multi-layer perceptron as the final model.

## III. Proposed Methodology

The proposed model consists of data collection, feature extraction, ensemble learning-based prediction, and the application of explainable AI. The dataset contains 651,191 URLs, which are classified as benign (or safe URL), defacement, phishing, or malware. 21 features are extracted from these URL strings, such as the count of special characters such as '.', '@', '?', URL length, count of digits, etc. Table II shows their short forms and their descriptions. Then 70% of these URL features are fed to the stacked ensemble learning model for training, and the rest 30% are used for testing. The ensemble learning model comprises two levels - level 0 consists of LightGBM, Gradient boost, XGBoost, and random forest, and level 1 consists of a random forest model. Then Explainable AI (XAI) is used to explain the working of the model and demystify the black-box nature of machine learning algorithms. Figure 1 shows the flowchart explaining the proposed methodology, and x1, x2, etc. are each of the 21 features.

### A. Dataset and Feature Extraction

A dataset of 651,191 URLs was used to train the models. The dataset is available on kaggle.com and was collected from widely-used sources by researchers of malicious URL detection domains such as Phishtank [14] [15] (accessed on June 30, 2022) and URL dataset [11] (ISCX-URL-2016, accessed on June 30, 2022). The training data consists of a URL string that is labeled as one of the 4 classes - benign, defacement, phishing, or malware. Figure 2 shows the count of the 4 types of URLs in the dataset. The target variable - the URL classification, is based on the values of these 21 variables. Each URL is classified into one of the 4 categories.

*1) Benign:* Safe-to-click links (or URLs) are called benign URLs.

*2) Phishing:* Phishing URLs are used to obtain sensitive information such as passwords or login credentials, banking details, etc. They can often be sent in emails, or SMS, maybe after which after clicking malicious files can also be downloaded into the user's system.

*3) Defacement:* This is a type of website vandalism, where the attacker modifies the contents of the website mostly by 'cyber protestors' or 'hacktivists'. Businesses afflicted by defacement attacks face loss of customer loss.

*4) Malware:* A malware attack is where malware or malicious software performs unauthorized actions on the victim's system, to harm a computer, server, client, or computer network. Then, data is stolen, the system is damaged, the system is monitored, etc. without the user knowing.

### B. Machine Learning Models

This paper treats the classification of URLs as a multi-class problem, which answers the need that different attacks have distinctive levels of threat. This approach helps to identify varying threats more than just classifying URLs as safe or unsafe, and then take attack-specific protection for each classified case. Many machine learning algorithms have been used



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TABLE I
COMPARISON OF THIS WORK WITH RELATED WORKS

| Author | Topic | Main feature | ML | XAI |
|---|---|---|---|---|
| Sheng et al. [9] | Empirical analysis of phishing blacklists | Exhaustive URL blacklist can't be maintained, given the ease of new malicious URL generation | No | No |
| Hou et al. [10] | Malicious web content detection by machine learning | Usage of lexical features of HTML and JS in web pages to detect malicious content with machine learning | Yes | No |
| Raja et al. [11] | Lexical features based malicious URL detection using machine learning techniques | URL classification as malicious or benign using various ML models - SVC, Logistic Regression, kNN, Naive Bayes, random forest | Yes | No |
| Patil et al. [12] | Detecting malicious URLs using decision trees | Malicious URLs detection using Decision Tree Classifiers and Majority Voting technique | Yes | No |
| Ghaleb et al. [13] | Threat detection using ensemble learning | Use of Random Forest and Multi-Layer perceptron in an ensemble model | Yes | No |
| This work | Malicious URL detection using ensemble learning and usage of Explainable AI to increase model interpretability | Ensemble of gradient boosting frameworks and Random Forest used in ensemble model. XAI used to show impact of each of the 21 features on the output | Yes | Yes |

TABLE II
LIST OF FEATURES

| SL No. | Features | Feature Description |
|---|---|---|
| 1 | use_of_ip | If IP is present in URL or not (boolean: 0 or 1) |
| 2 | abnormal_url | If the URL has an abnormal host name or not (boolean: 0 or 1) |
| 3 | count. | Count of dot symbol (.) (number) |
| 4 | count-www | Count of 'www' in the URL (number) |
| 5 | count@ | Count of '@' in the URL (number) |
| 6 | count_dir | Count of paths or number of directories in the URL (number) |
| 7 | count_embed_domian | Count of embedded links (number) |
| 8 | short_url | If URL is shortened or not (boolean: 0 or 1) |
| 9 | count-https | Count of 'https' in the URL (number) |
| 10 | count-http | Count of 'http' in the URL (number) |
| 11 | count% | Count of % symbols in the URL (number) |
| 12 | count? | Count of ? symbols in the URL (number) |
| 13 | count- | Count of - symbols in the URL (number) |
| 14 | count= | Count of = symbols in the URL (number) |
| 15 | url_length | Length of the URL (number) |
| 16 | hostname_length | Length of the host name (number) |
| 17 | sus_url | If the URL contains any suspicious words or not (boolean: 0 or 1) (such as bank, account, update, free, service, bonus, etc.) |
| 18 | fd_length | Length of first directory (number) |
| 19 | tld_length | Length of top-level domain (number) |
| 20 | count-digits | Count of digits in the URL (number) |
| 21 | count-letters | Count of letters in the URL (number) |

in literature, which can be used for malicious URL detection. However, after comparing the results of many algorithms, two approaches have been listed for this classification task.

*1) **Ensemble Machine Learning**:* An ensemble of models is a single model that uses predictions from multiple machine learning models to produce a model with better performance and predictability. In contrast to an ordinary machine learning model, which establishes a mapping between input and output by generating a relation function, stacked ensemble models learn the relation between the prediction results of each model used in the ensemble, to the actual class value.

The framework of a stacked ensemble model comprises two or more base models (level-0 models) and a higher-level meta-model (level-1). The level-0 or base models are models that fit the training data and predict out-of-sample data. The higher level model is a model that fits the prediction from the base models and learns the best combinations of the predictions. In this paper, we use *lightGBM*, *gradient boost*, *XGBoost*, and *random forest* as the base or level-0 models, and *random forest, with 5 cross folds* as the meta-model or the final estimator. Stacked ensembles produce more robust predictions than individual models or average ensembles, and they can optimize the best combinations of models.

- **LightGBM (Light Gradient Boosting Machine):** is a gradient boosting architecture based on decision trees that betters the performance of a model and reduces the usage of memory.
- **Gradient Boosting:** is an iterative boosting algorithm that minimizes a loss function by repeatedly choosing a function towards the negative gradient, or simply, each predictor corrects the errors of its predecessor. Weak learners, which are usually decision trees, are added sequentially step by step, to reduce the value of the loss function.
- **XGBoost (eXtreme Gradient Boosting):** is decision trees (DT) created in a sequential manner. Weights are



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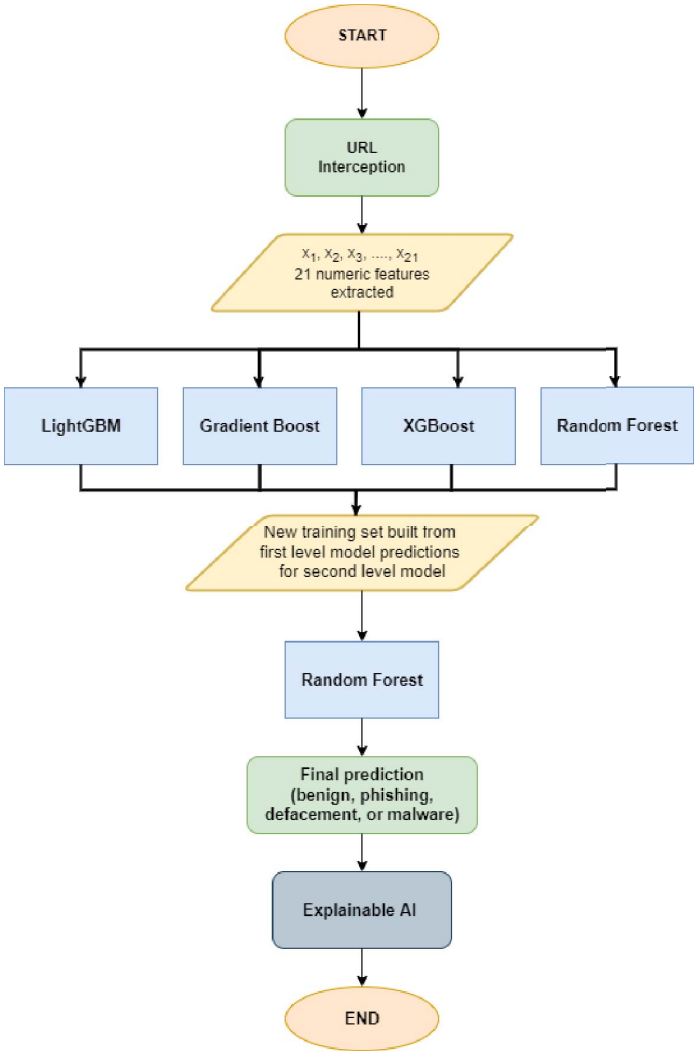



Fig. 1. Flowchart of Proposed Methodology

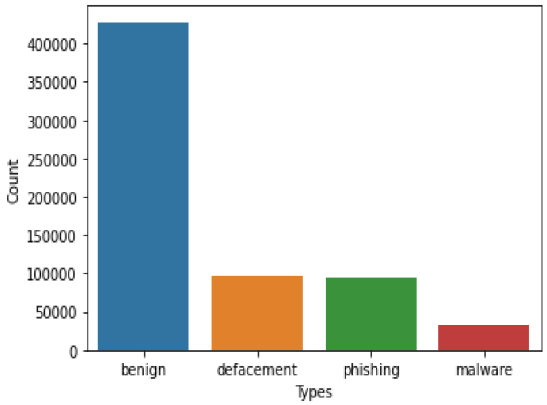



Fig. 2. Count of different URL classes

assigned to all independent variables which are fed to the DT, which predicts the results. The weight of the DT which was wrong is then increased, and these variables are then fed to the next DT.

- **Random Forest:** is a supervised learning algorithm, which makes use of an ensemble of decision trees, which leads to more accurate results.

### *C. Explainable AI*

Machine Learning models are essentially a black box due to the poor explainability of the working of the models. Explainable AI (XAI) is used to explain the working of the model and hence, increase the interpretability of the model. The application of XAI has been clearly shown in figure 4. Simply put, XAI helps demonstrate how the predictions of the model will change if the value of a single feature is changed.



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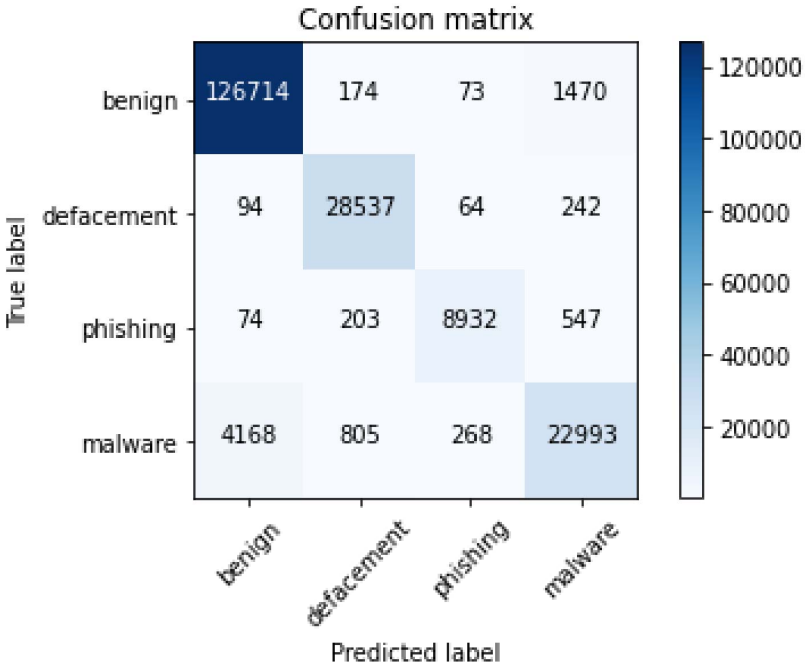



Fig. 3. Confusion Matrix of Ensemble ML model

The usage of SHAP values has been used to implement XAI in a classification problem [16], and its implementation has been shown clearly. However, ALE values have been used to implement XAI in this paper.

- **ALE:** Accumulated Local Effects (ALE) [17] calculates first-order feature effects on the model with respect to a dataset. It is a method for computing feature effects on the paper [18]. ALE plots are an extension of partial dependence plots (PDPs). PDP works by marginalizing the model's output over the features we are not interested in (say X). The process of factoring out the set X causes the introduction of artificial data, which can lead to errors. ALE plots solve this by using the conditional probability distribution instead of the marginal distribution and removing any incorrect output dependencies due to correlated input variables by accumulating local differences in the model output instead of averaging them.

## IV. Performance Evaluation

The proposed model discussed in this paper has been validated using the above-mentioned dataset which is classified into 4 types of malicious URLs, and performance measures. 21 features have been extracted from the dataset and then 70% of these URLs were fed to the stacked ensemble learning model for training, and the rest 30% are used for testing. Further, Explainable AI has been used to explain the working of the model, or the impact of each feature on the model prediction has been demonstrated.

Table III shows the class-wise performance metrics of the ensemble learning model - level 0 of the model consists of LightGBM, Gradient boost, XGBoost, and random forest, and level 1 consists of random forest model. As seen from the table, the accuracy of the model is 97%.

TABLE III
Class-wise Performance metrics of the Ensemble ML model

| Class | Precision | Recall | f1-score | Support |
|---|---|---|---|---|
| 0(benign) | 0.97 | 0.99 | 0.98 | 128431 |
| 1(defacement) | 0.98 | 0.99 | 0.99 | 28937 |
| 2(phishing) | 0.99 | 0.94 | 0.96 | 9756 |
| 3(malware) | 0.91 | 0.86 | 0.89 | 28234 |
| **Accuracy** | | | **0.97** | 195358 |
| Macro Average | 0.96 | 0.95 | 0.95 | 195358 |
| Weighted Average | 0.97 | 0.97 | 0.97 | 195358 |

Figure 3 shows the confusion matrix of the ensemble learning classifier (level 0 - LightGBM, Gradient boost, XGBoost, random forest, and level 1 - random forest) which conveys the information of true prediction and false prediction by the model.

Table IV shows the individual model performances. It is clear from the table that the ensemble model which uses all of the other stated models, has higher accuracy and lesser Root Mean Squared Error (RMSE) than any of the individual models.

Instead of one figure to explain the model, 21 different plots have been demonstrated to explain the effect of each of the 21 features on the model prediction of each class. Figure 4



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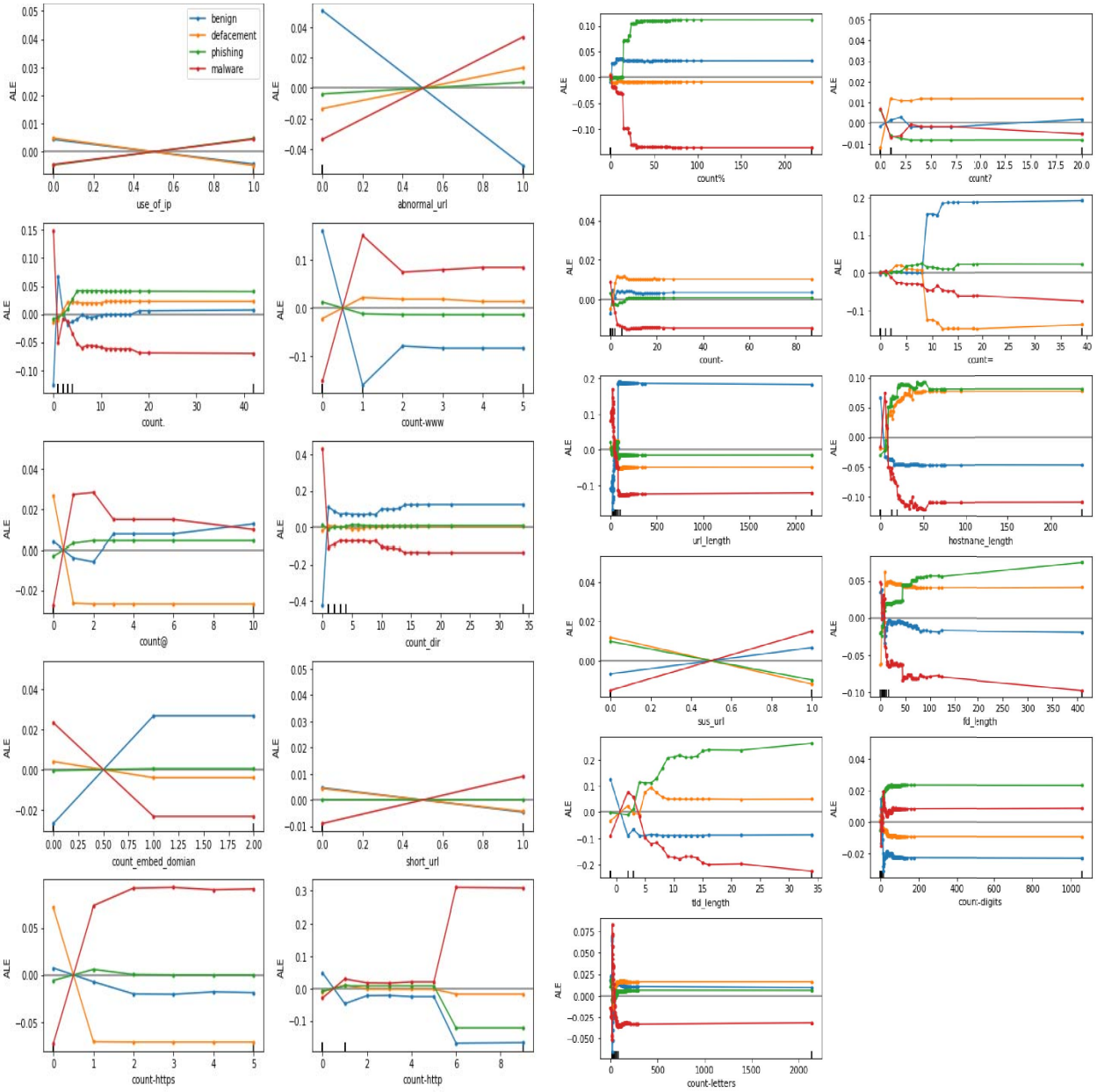



Fig. 4. Explainable AI using ALE

shows the ALE (Accumulated Local Effects) plots explaining the feature effects on the probabilities of each class. The high non-linearity of the plots shows that a non-linear model was used for prediction, which wouldn't be in the case of a linear model. In this case, the ALE is in the units of relative probability mass, i.e. given a feature value how much more (or less) probability does the model assign to each class relative to the mean prediction. This also means that any increase in relative probability of one class must result in a decrease in the probability of another class. The ALE curves summed across classes result in 0 as a direct consequence of the conservation of probability.

In figure 4, the x-axis values in each of the plots represent the range of values of each of the 21 features (table II shows all the feature details), for example, in plot 1, in use_of_ip, which is a boolean value (0 or 1), it can be noted that there are points only in 0 and 1, corresponding to each point in the plot's y-axis is the feature effect on the final prediction. The



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TABLE IV
INDIVIDUAL MODEL PERFORMANCE

| Model | Test Accuracy | RMSE |
|---|---|---|
| LightGBM | 95.62% | 0.55 |
| Gradient Boost | 95.78% | 0.54 |
| XGBoost | 93.37% | 0.63 |
| Random Forest | 96.58% | 0.51 |
| Ensemble (this model) | 97% | 0.50 |

same goes for all the other boolean-valued features (sus_url, abnormal_url, etc.). Similarly, in continuous-valued features like 'count.' (in which the x-axis values range from 0 to 43, 0 to 5 for 'count-www', and so on) each point in the plot shows its respective effect on the final prediction. The same goes for other continuous-valued features, such as 'count-', 'url_length', etc.

It can be noted that in the plot for count-embed-domain, for phishing, the ALE plot is virtually flat, which means the model does not use this feature to assign a higher or lower probability to class phishing with respect to the average prediction, whereas, in the count-https figure, the model assigns higher probability to malware with respect to the count-https feature, and lower probability for defacement for the count-https feature. Hence, the use of Explainable AI makes it very clear exactly how much each feature is impacting the result of the ML model.

## V. CONCLUSIONS AND FUTURE WORK

Malicious URL detection remains the most common type of attack on users, and so their detection has been an important field of study for all cyber-security specialists. With the day-to-day increase in the use of the internet and an increase in online businesses and services, cyberattacks are only going to increase. Thus it becomes very important to research different methods to prevent and defend systems and users against harmful attacks. Due to the diversity of attack vectors of malicious websites, 21 features were extracted from 650,000 to train the proposed model. The second contribution of this work is in the proposal of a two-stage stacked ensemble learning model, based on gradient boosting methods and random forest, which were all trained and tested in the 70:30 ratio of such a large number of URLs, and an accuracy of 97% has been achieved. The third main contribution of the model is the use of Explainable AI, which has not been used in any other literature related to this field of study, to explain the working of the model, and study the impact of each of the 21 features on the 4 class predictions. Future works can be done to increase the accuracy of the model, and a newer or updated dataset can be used to train the model, as malicious URLs are generated every day, so web crawling can be used to perform live predictions. Further, applications of Explainable AI can be done in various other fields of research, and it can be used to achieve better results in different use cases, as explaining the effects of each of the features on the model predictions becomes very easy, and clear to understand. The paper can be further integrated with Blockchain and framework like AIBlock can be designed [19].

## VI. DATA AVAILABILITY STATEMENT

This study used a malicious URLs dataset, publicly available as Malicious URLs dataset on kaggle.com (accessed on June 30, 2022).

## REFERENCES


[1] A. Abubakar Aliyu, B. H. Tasmin *et al.*, "The impact of information and communication technology on banks performance and customer service delivery in the banking industry," 2012.

[2] T. N. Jagatic, N. A. Johnson, M. Jakobsson, and F. Menczer, "Social phishing," *Communications of the ACM*, vol. 50, no. 10, pp. 94–100, 2007.

[3] A. K. Sood and S. Zeadally, "Drive-by download attacks: A comparative study," *It Professional*, vol. 18, no. 5, pp. 18–25, 2016.

[4] S. Gupta, A. Singhal, and A. Kapoor, "A literature survey on social engineering attacks: Phishing attack," in *2016 international conference on computing, communication and automation (ICCCA)*. IEEE, 2016, pp. 537–540.

[5] Z. Su and G. Wassermann, "The essence of command injection attacks in web applications," *Acm Sigplan Notices*, vol. 41, no. 1, pp. 372–382, 2006.

[6] T. Berners-Lee, L. Masinter, and M. McCahill, "Uniform resource locators (url)," Tech. Rep., 1994.

[7] D. Sahoo, C. Liu, and S. C. Hoi, "Malicious url detection using machine learning: A survey," *arXiv preprint arXiv:1701.07179*, 2017.

[8] F. Vanhoenshoven, G. Nápoles, R. Falcon, K. Vanhoof, and M. Köppen, "Detecting malicious urls using machine learning techniques," in *2016 IEEE Symposium Series on Computational Intelligence (SSCI)*. IEEE, 2016, pp. 1–8.

[9] S. Sheng, B. Wardman, G. Warner, L. Cranor, J. Hong, and C. Zhang, "An empirical analysis of phishing blacklists," 2009.

[10] Y.-T. Hou, Y. Chang, T. Chen, C.-S. Laih, and C.-M. Chen, "Malicious web content detection by machine learning," *expert systems with applications*, vol. 37, no. 1, pp. 55–60, 2010.

[11] A. S. Raja, R. Vinodini, and A. Kavitha, "Lexical features based malicious url detection using machine learning techniques," *Materials Today: Proceedings*, vol. 47, pp. 163–166, 2021.

[12] D. R. Patil, J. B. Patil *et al.*, "Malicious urls detection using decision tree classifiers and majority voting technique," *Cybernetics and Information Technologies*, vol. 18, no. 1, pp. 11–29, 2018.

[13] F. A. Ghaleb, M. Alsaedi, F. Saeed, J. Ahmad, and M. Alasli, "Cyber threat intelligence-based malicious url detection model using ensemble learning," *Sensors*, vol. 22, no. 9, p. 3373, 2022.

[14] S. Bell and P. Komisarczuk, "An analysis of phishing blacklists: Google safe browsing, openphish, and phishtank," in *Proceedings of the Australasian Computer Science Week Multiconference*, 2020, pp. 1–11.

[15] S. Marchal, J. François, R. State, and T. Engel, "Phishstorm: Detecting phishing with streaming analytics," *IEEE Transactions on Network and Service Management*, vol. 11, no. 4, pp. 458–471, 2014.

[16] D. Chowdhury, S. Poddar, S. Banarjee, R. Pal, A. Gani, C. Ellis, R. C. Arya, S. S. Gill, and S. Uhlig, "Covidxai: Explainable ai assisted web application for covid-19 vaccine prioritisation," *Internet Technology Letters*, p. e381, 2022.

[17] J. Klaise, A. V. Looveren, G. Vacanti, and A. Coca, "Alibi explain: Algorithms for explaining machine learning models," *Journal of Machine Learning Research*, vol. 22, no. 181, pp. 1–7, 2021. [Online]. Available: http://jmlr.org/papers/v22/21-0017.html

[18] D. W. Apley and J. Zhu, "Visualizing the effects of predictor variables in black box supervised learning models," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 82, no. 4, pp. 1059–1086, 2020.

[19] M. Golec, D. Chowdhury, S. Jaglan, S. S. Gill, and S. Uhlig, "Aiblock: Blockchain based lightweight framework for serverless computing using ai," in *2022 22nd IEEE International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*. IEEE, 2022, pp. 886–892.




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